8.3 Spacer

All couplings shall be of the spacer type. The spacer shall be of sufficient length to allow removal of coupling hubs and to allow for maintenance of adjacent bearings and seals without removal of the shaft or disturbance of the equipment alignment. Unless otherwise specified by the purchaser, the spacer length shall correspond to a distance between shaft-ends of 460 mm (18 in).

NOTE The spacer gap length is not necessarily the same as the spacer length.

8.4 Hub type

- The purchaser shall specify whether the coupling shall be designed to fit to integral flange(s) or be provided with removable hub(s).

NOTE The two ends of the couplings are not necessarily the same.

8.5 Integral flanges

8.5.1 If the coupling is to fit to integrally flanged shaft end(s), the coupling vendor and the manufacturer with unit responsibility shall agree on the flange geometry. The holes shall be machined with computer numerical-controlled (CNC) equipment. A drill fixture (or template) shall be used only if CNC equipment is not available. In either case, inspection shall be performed to assure proper location of the mating coupling flanges. The coupling vendor shall supply the drill fixture (or template), if required.

8.5.2 Unless otherwise specified, the coupling shall mate directly to the integral flange without an adapter.

NOTE 1 Addition of an adapter between the coupling and the integral flange increases the overhung moment and adds another fit whose runout and clearance it is necessary to control in order to maintain balance. Further, the issue of who is responsible for supplying the adapter (machine manufacturer or coupling manufacturer) requires an agreement for each order.

NOTE 2 For certain machines, particularly gas turbines, space limitations can make it impractical to comply with the requirement in 8.5.2.

8.6 Hubs

8.6.1 Removable hubs

- **8.6.1.1** Removable coupling hubs shall be fitted to the shaft using an interference fit to positively locate and centre the hub, and therefore the rest of the coupling, on the shaft. These hubs may have a parallel or taper bore, with or without keyways, or may be splined with a major diameter fit, or splined with a side fit and fore and aft locating pilots, as specified by the purchaser.

- The degree of interference shall be specified by the purchaser and is subject to approval by the vendor.

The following guidelines are recommended for hub to shaft fits.

a) The interference fit for straight-bore, keyed hubs should be from 0.000 50 mm/mm (in/in) to 0.000 75 mm/mm (in/in) of bore diameter.

b) The interference fit for tapered-bore, keyed hubs should be at least 0.001 mm/mm (in/in) of bore diameter.

c) The interference rate for taper-bore, hydraulically fitted hubs should be based upon the torque that is required to be transmitted. As a maximum, the interference rate should not exceed 0.003 mm/mm (in/in) of bore diameter to avoid potential bonding problems.

d) Some international machinery manufacturers may use preferred fits for shafts and bores.

- **8.6.1.2** If specified, for straight-bore, keyed hubs, shaft sizes and coupling bores shall be in accordance with ANSI/AGMA 9002, ANSI/AGMA 9112 or ISO 286-2.

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